

5.6 Reaction Kinetics (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	5. Physical Chemistry (A Level Only)
Topic	5.6 Reaction Kinetics (A Level Only)
Difficulty	Hard

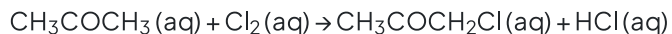
Time allowed: 40

Score: /30

Percentage: /100

Question 1a

In the presence of acid, $\text{H}^+(\text{aq})$, chlorine and propanone react together:



A student carried out an investigation into the kinetics of this reaction.

The student investigated how different concentrations of chlorine affect the initial rate of the reaction. A graph of $[\text{Cl}_2(\text{aq})]$ against time is shown in Fig. 1.1.

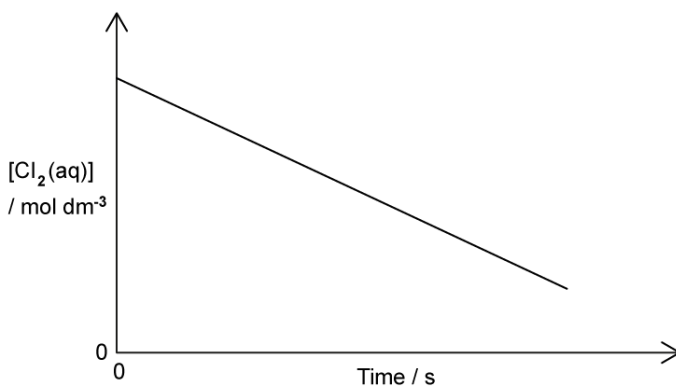


Fig. 1.1

The student then investigated how different concentrations of propanone and $\text{H}^+(\text{aq})$ affect the initial rate of reaction.

Their results are shown below.

Experiment	$[\text{Cl}_2(\text{aq})]$ / mol dm^{-3}	$[\text{CH}_3\text{COCH}_3(\text{aq})]$ / mol dm^{-3}	$[\text{H}^+(\text{aq})]$ / mol dm^{-3}	Initial rate / $\text{mol dm}^{-3} \text{s}^{-1}$
1	0.003	0.75	0.05	0.18×10^{-5}
2	0.003	1.50	0.15	1.08×10^{-5}
3	0.003	1.50	0.30	2.16×10^{-5}

a)

Use the student's results to determine the reaction orders. Explain your answer.

[4 marks]

Question 1b

Deduce the rate equation and calculate the rate constant for this reaction, including the units.

Rate equation

$k = \dots\dots\dots$

units =

[4 marks]

Question 1c

The student varied the concentrations of the reactants in another experiment and they found the rate to be $0.26 \times 10^{-5} \text{ mol dm}^{-3} \text{ s}^{-1}$. They also used a pH probe and found that the reaction mixture had a pH of 2.

Calculate the initial rate of the reaction mixture if the amount of acid added was altered to give a pH of 1.

Assume the temperature and the initial concentrations of the other reactants remained the same.

Initial rate =

[1 mark]

Question 1d

The experiment was repeated at a lower temperature. State what the effect would be of this change, if any, on the rate and the rate constant of the reaction.

[2 marks]**Question 2a**

Nitrogen monoxide, NO, reacts with oxygen, O₂. The rate equation for this reaction is shown below.

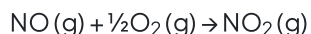
$$\text{rate} = k[\text{NO}(\text{g})]^2[\text{O}_2(\text{g})]$$

State what would happen to the rate of the reaction if the initial concentration of NO is tripled and the initial concentration of O₂ is halved.

[1 mark]**Question 2b**

Nitrogen monoxide is produced by combustion in car engines and released into the atmosphere.

In the lower atmosphere, nitrogen monoxide is responsible for the formation of ozone in a series of reactions shown below.



i)

What is the overall equation for this series of reactions?

[1]

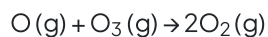
ii)

Explain why NO is acting as a homogenous catalyst in this reaction.

[2]**[2 marks]**

Question 2c

When nitrogen monoxide is present in the upper atmosphere, is it involved in the removal of ozone, O_3 . The overall equation for the reaction is shown below.



The rate equation is: $\text{rate} = k[\text{NO}(\text{g})][\text{O}_3(\text{g})]$

i)

State how this shows that the mechanism contains more than one step.

[1]

ii)

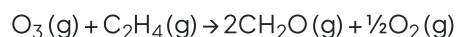
Deduce a possible two-step mechanism for this reaction.

[2]

[3 marks]

Question 2d

Ozone in the lower atmosphere can react with ethene to produce methanal, $\text{CH}_2\text{O}(\text{g})$ which contributes to low-level smog. The equation is shown below.



The order of the reaction with respect to both reactants is first order. The initial rate of formation of methanal was found to be $2.0 \times 10^{-11} \text{ mol dm}^{-3} \text{ s}^{-1}$.

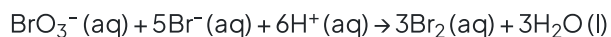
The initial concentration of ozone was doubled and the initial concentration of ethene was tripled.

Calculate the initial rate of oxygen formation.

[2 marks]

Question 3a

Bromate(V) ions and bromide ions react in the presence of acid to form bromine and water.



A student carried out a series of experiments to investigate how the rate of the reaction of bromate and bromide in the presence of an acid varies with temperature.

The time taken, t , was measured for a fixed amount of bromine to form at different temperatures. The results are shown in the Table 3.1.

Table 3.1

Temperature (T) / K	$\frac{1}{T} \times 10^{-3} / \text{K}^{-1}$	Time (t) / s	$\frac{1}{t} / \text{s}^{-1}$	$\ln \frac{1}{t}$
408	2.451	21.14	0.0473	-3.051
428		10.57	0.0946	-2.358
448	2.232	5.54		
468	2.137			-1.106
488	2.049	1.71	0.5851	-0.536

Calculate the missing values to complete the table above.

[3 marks]

Question 3b

The relationship between the rate constant, k , and the activation energy, E_a , and temperature, T is given by the following equation.

$$\ln k = \ln A + \frac{-E_a}{RT}$$

In this experiment, the rate constant, k , is directly proportional to $\frac{1}{t}$. Therefore,

$$\ln \frac{1}{t} = \ln A + \frac{-E_a}{RT}$$

Use your answers from part (a) to plot a graph of $\ln \frac{1}{t}$ against $\frac{1}{T} \times 10^{-3}$ on the graph in Fig. 3.1.

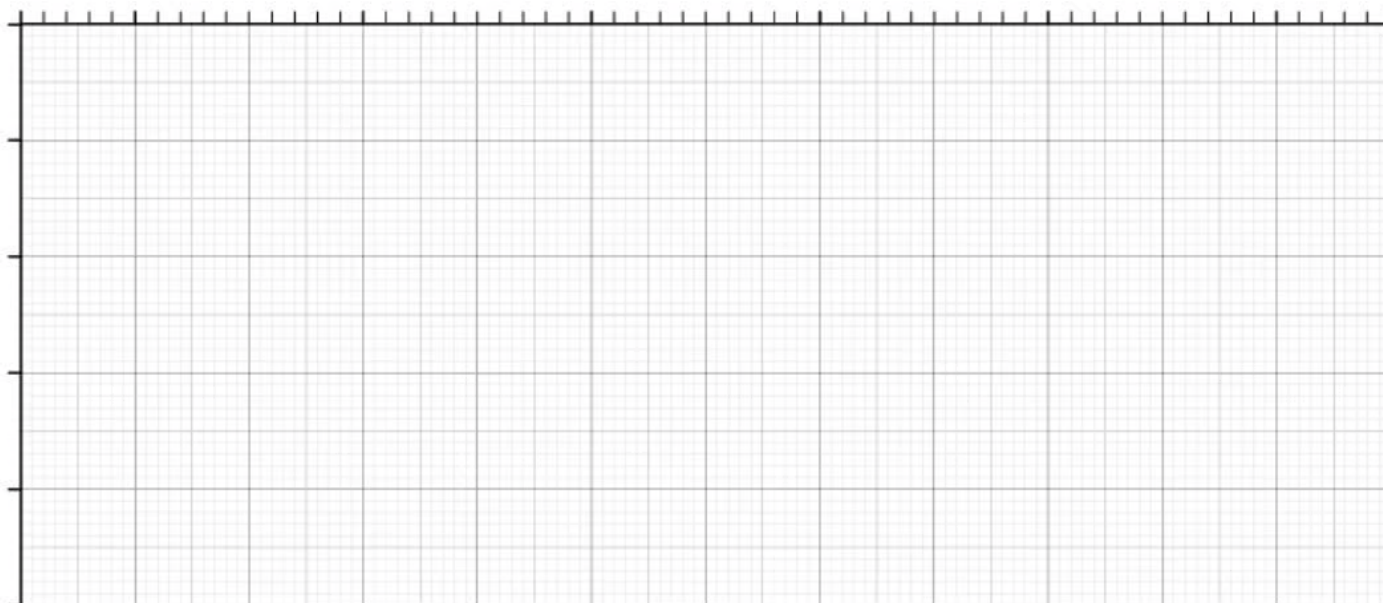


Fig. 3.1

[4 marks]

Question 3c

Use your graph in Fig. 3.1 and information from part (c) to calculate a value for the activation energy, in kJ mol^{-1} , for this reaction. Show your working.

The value of the gas constant, $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$

$E_a = \dots\dots\dots \text{kJ mol}^{-1}$

[4 marks]

Model Answers: Hard

1a

a) Using the student's results shows that the reaction orders are:

- Zero / 0 order with respect to Cl_2

AND

Constant gradient / linear negative gradient / constant rate / rate independent of concentration; [1 mark]

- First / 1st order with respect to H^+

AND

From experiment 2 to 3, $[\text{H}^+(\text{aq})]$ doubles / x 2 and rates doubles / x 2; [1 mark]

- First / 1st order with respect to propanone / CH_3COCH_3 ; [1 mark]

- From experiment 1 to 2, $[\text{CH}_3\text{COCH}_3(\text{aq})]$ x 2 **AND** $[\text{H}^+(\text{aq})]$ x 3 **AND** rate x 6

OR

From experiment 1 to 3, $[\text{CH}_3\text{COCH}_3(\text{aq})]$ x 2 **AND** $[\text{H}^+(\text{aq})]$ x 6 **AND** rate x 12; [1 mark]

[Total: 4 marks]

- You need to be able to determine the order with respect to a reactant from the shape of a concentration-time graph
 - A zero order reaction will produce a straight line with a negative gradient
 - The gradient is constant and the value is equal to the rate constant, k
- You need to work out the order with respect to H^+ first as there is not a pair of experiments in which the concentration of CH_3COCH_3 changes while the concentrations of both Cl_2 and H^+ remain the same
- From experiment 2 to 3 only $[\text{H}^+(\text{aq})]$ changes, as $[\text{H}^+(\text{aq})]$ doubles, so does the rate
 - This means it must be first order with respect to H^+ as the rate is directly proportional to the concentration
- Now you have deduced the orders with respect to Cl_2 and H^+ , you can use the results to work out the order with respect to CH_3COCH_3
 - Between experiments 1 and 2, the rate increased 6 times
 - 3 times due to $[\text{H}^+(\text{aq})]$ tripling, and therefore 2 times again due to



- This was caused by a doubling of $[\text{CH}_3\text{COCH}_3(\text{aq})]$, and so must be first order with respect to CH_3COCH_3

OR

- Between experiments 1 and 3, the rate increased 12 times
- 6 times due to $[\text{H}^+(\text{aq})]$ increasing 6 times, and therefore 2 times again due to CH_3COCH_3
- This was caused by a doubling of $[\text{CH}_3\text{COCH}_3(\text{aq})]$, and so must be first order with respect to CH_3COCH_3

1b

b) The rate equation and rate constant for this reaction, including the units are:

- $\text{rate} = k[\text{CH}_3\text{COCH}_3(\text{aq})][\text{H}^+(\text{aq})]$; [1 mark]
- $k = \frac{\text{rate}}{[\text{CH}_3\text{COCH}_3(\text{aq})][\text{H}^+(\text{aq})]}$

OR

- $\frac{0.18 \times 10^{-5}}{(0.75) \times (0.05)}$; [1 mark]
- $= 4.80 \times 10^{-5}$; [1 mark]
- Units: $\text{dm}^3 \text{mol}^{-1} \text{s}^{-1}$; [1 mark]

[Total: 4 marks]

- $[\text{Cl}_2(\text{aq})]$ will not appear in the rate equation as it is zero order with respect to Cl_2
- Both $[\text{CH}_3\text{COCH}_3(\text{aq})]$ and $[\text{H}^+(\text{aq})]$ will be raised to the power of 1 as the reaction is first order with respect to both of these reactants
- **Careful:** It is a common error for students to rearrange the rate equation incorrectly, if you are not confident with rearranging, it can be easier to substitute the values into the rate equation first, simplify and then rearrange
- You can check your answer by using your calculated rate constant and some values from the other experiments to see if you calculate the rate correctly
- **Note:** You would be awarded marks if you based your calculations on any of the experiments – they all give the same value of k
 - It is the convention to write positive indices before negative indices but if you gave the units as $\text{mol}^{-1} \text{dm}^3 \text{s}^{-1}$ you would still be awarded the mark
 - You would be given 'error carried forward' marks if you obtained the

incorrect rate equation but the remainder of your calculations based on this were correct

1c

c) The initial rate of the reaction mixture if the amount of acid added was altered to give a pH of 1 would be:

- Initial rate = $0.26 \times 10^{-5} \times 10 = 2.60 \times 10^{-5} \text{ (mol dm}^{-3} \text{ s}^{-1}\text{)}$; [1 mark]

[Total: 1 mark]

- This question tests your understanding of pH and the relationship between rates and order
- The concentration of H^+ would be 10 times greater at a pH of 1 than at a pH of 2
- As the reaction is first order with respect to H^+ , the initial rate will be 10 times greater at pH 1 than at pH 2
- A common mistake would be to divide by 10; think carefully, the lower the pH, the greater the concentration of H^+ which must mean a faster rate of reaction

1d

d) The effect of lowering temperature on the rate and the rate constant of the reaction would be:

- The rate decreases; [1 mark]
- The rate constant / k decreases; [1 mark]

[Total: 2 marks]

- An increase in temperature will increase both the rate of reaction and the rate constant
- Typically, for every 10°C increase in temperature, the rate and the rate constant will double

2a

a) If the initial concentration of NO is tripled and the initial concentration of O_2 is halved, the rate of reaction would:

- Increase by a factor of 4.5; [1 mark]

[Total: 1 mark]

- The reaction is 2nd order with respect to NO, therefore the rate would change by a factor of $3^2 = 9$ as the initial concentration of NO tripled
- The reaction is 1st order with respect to O₂, therefore the rate would half as the initial concentration of O₂ halved, i.e. it changes by a factor of $\frac{1}{2}$
- To find the combined effect on the rate of these changes, multiply these factors together: $9 \times \frac{1}{2} = 4.5$ therefore the reaction rate would increase by a factor of 4.5

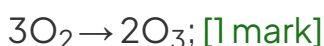
2b

b)

i) The overall equation for this series of reactions is:



OR



ii) NO is acting as a homogenous catalyst because:

- It is in the same phase as the reaction mixture; [1 mark]
- It is (used in the first step and) regenerated in the second step; [1 mark]

[Total: 2 marks]

- To find the overall equation put all of the reactants together and all of the products together:
 - $\text{NO}(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) + \text{NO}_2(\text{g}) + \text{O}_2(\text{g}) + \text{O}(\text{g}) \rightarrow \text{NO}_2(\text{g}) + \text{NO}(\text{g}) + \text{O}(\text{g}) + \text{O}_3(\text{g})$
- Cancel out any species that appear on both sides of the equation and simplify:
 - ~~$\text{NO}(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) + \text{NO}_2(\text{g}) + \text{O}_2(\text{g}) + \text{O}(\text{g}) \rightarrow \text{NO}_2(\text{g}) + \text{NO}(\text{g}) + \text{O}(\text{g}) + \text{O}_3(\text{g})$~~
 - $\frac{1}{2}\text{O}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow \text{O}_3(\text{g})$
 - $1\frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{O}_3(\text{g})$
- You can leave the equation with fractions or multiply by 2 to give whole numbers
 - $3\text{O}_2(\text{g}) \rightarrow 2\text{O}_3(\text{g})$
- You do not need to give the state symbols

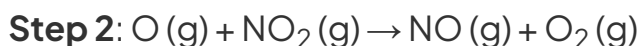
2c

c)

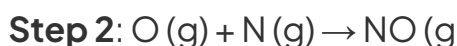
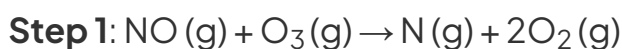
i) The overall equation and rate equation shows that the mechanism contains more than one step because:

- The overall equation does not match the rate equation; [1 mark]

ii) Possible two-step mechanisms for this reaction are:



OR



- Correct reactants in step 1; [1 mark]
- Correct products in step 1 and correct reactants and products in step 2; [1 mark]

[Total: 3 marks]

- You do not need to give the state symbols to be awarded the marks
- The 2nd mark can only be awarded if the 1st mark has been awarded AND step 1 and step 2 add up to the overall equation
- You could have given alternative products in step 1 and reactants and products in step 2 provided that step 1 and step 2 add up to the overall equation
- The rate equation only includes the reacting species up to the rate-determining step
- As this is the first step, the reactants must be only those that appear in the rate equation and they are both first order so only one of each species will be involved in the rate-determining step:
 - $\text{NO (g)} + \text{O}_3\text{(g)}$
- These could react to form 2O_2 , leaving N left as an intermediate which would go on to react in the 2nd step
- Comparing the reactants from the rate-determining step with the overall equation, we can see that an atom of oxygen also needs to be used as a reactant, so the reactants for step 2 must be:
 - $\text{N (g)} + \text{O (g)}$

- The rate equation only includes the reacting species up to the rate-determining step
- As this is the first step, the reactants must be only those that appear in the rate equation and they are both first order so only one of each species will be involved in the rate-determining step:
 - $\text{NO (g)} + \text{O}_3 \text{ (g)}$
- These could react to form 2O_2 , leaving N left as an intermediate which would go on to react in the 2nd step
- Comparing the reactants from the rate-determining step with the overall equation, we can see that an atom of oxygen also needs to be used as a reactant, so the reactants for step 2 must be:
 - $\text{N (g)} + \text{O (g)}$
- All of the products in the overall equation have been accounted for so the product that these form will not be seen in the overall equation, the product must be NO (g) which was used in step 1 and is regenerated in step 2
- An alternative (more likely) two step mechanism is also given credit in which just one oxygen molecule is produced in step 1 and NO_2 is the intermediate product
- When writing possible mechanisms, you will not be penalised if the intermediate product is not viable or likely
 - As long the intermediate is used up in the next step and the sum of the steps in the mechanism matches the overall equation you will be awarded the marks

2d

d) The initial rate of oxygen formation is:

- Initial rate of methanol formation = $2.0 \times 10^{-11} \times 2 \times 3 = 1.2 \times 10^{-10} \text{ mol dm}^{-3} \text{ s}^{-1}$; [1 mark]
- Initial rate of oxygen formation = $1.2 \times 10^{-10} \div 4 = 3.0 \times 10^{-11} \text{ mol dm}^{-3} \text{ s}^{-1}$; [1 mark]

[Total: 2 marks]

- The reaction is first order with respect to both reactants so as the initial concentration of ozone doubled, the rate would double and as the initial concentration of ethene tripled, the rate would triple
- The initial rate of methanol formation is different from the initial rate of oxygen formation because for every 2 moles of methanol formed, 0.5 moles of oxygen are formed, so the rate of oxygen formation will be 4 times less than the rate of

methanal formation

3a

a) The completed table is:

Temperature (T) / K	$\frac{1}{T} \times 10^{-3} / \text{K}^{-1}$	Time (t) / s	$\frac{1}{t} / \text{s}^{-1}$	$\ln \frac{1}{t}$
408	2.451	21.14	0.0473	-3.051
428	2.336	10.57	0.0946	-2.358
448	2.232	5.54	0.1805	-1.712
468	2.137	3.02	0.3309	-1.106
488	2.049	1.71	0.5851	-0.536

- $\frac{1}{T} = \frac{1}{428} = 2.336 (\times 10^{-3}); [1 \text{ mark}]$
- $\frac{1}{t} = \frac{1}{5.54} = 0.1805$

AND

$$\ln \frac{1}{t} = \ln 0.1805 = -1.712 ; [1 \text{ mark}]$$

- $e^{\ln(1/t)} = e^{-1.106} = 0.3309$

AND

$$t = \frac{1}{0.3309} = 3.02; [1 \text{ mark}]$$

[Total: 3 marks]

- All of the missing values are completed on calculators
- To do, or reverse, the $\frac{1}{t}$ or $\frac{1}{T}$ calculation, you type $1 \div$ the corresponding value of t or T
 - e.g. $\frac{1}{0.331} = 3.02$
- To calculate $\ln \frac{1}{t}$ you press the $\ln$ button on your calculator and then type the corresponding value
 - e.g. $\ln 0.1805 = -1.712$
- To reverse the $\ln$ calculation, you press $e^{\square}$ or e^x on your calculator and then type

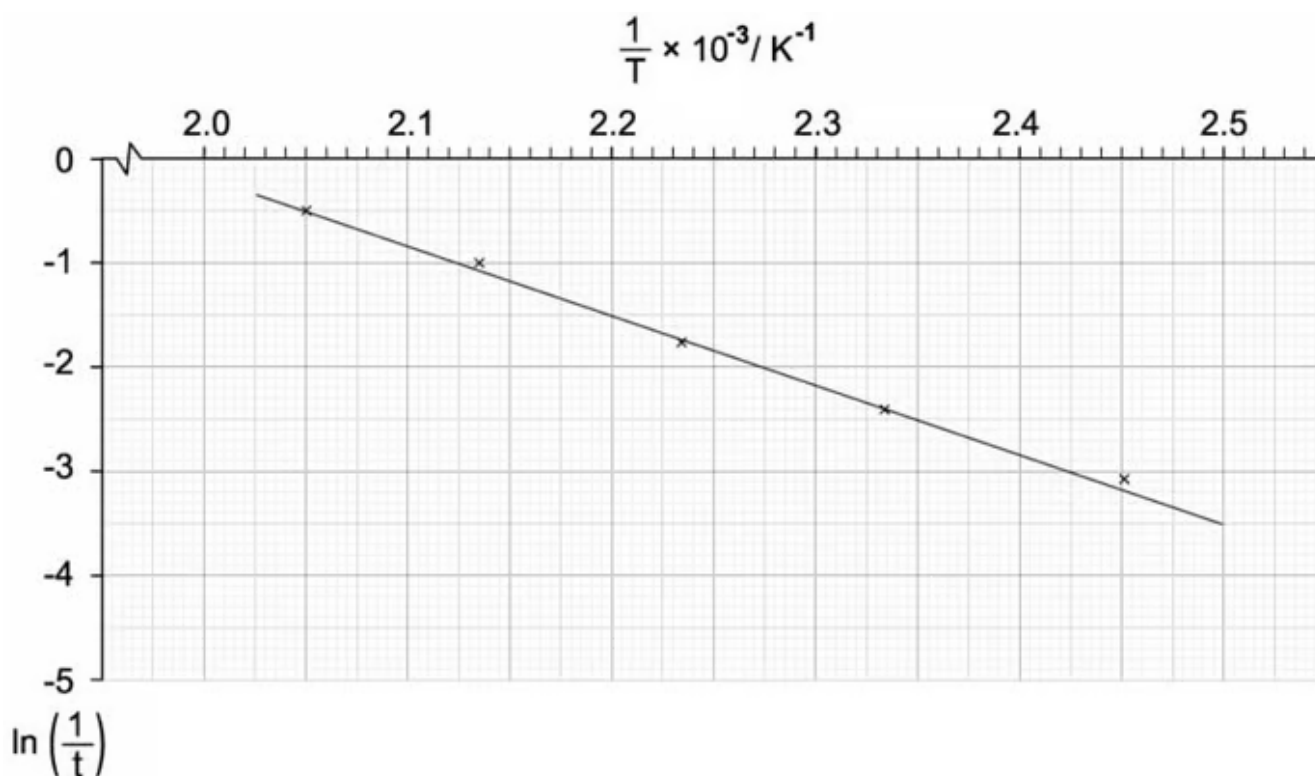
the corresponding value

o e.g. $e^{-1.106} = 0.3309$

- **Careful:** The heading for $\frac{1}{T}$ already gives $\times 10^{-3}$ so you do not need to write 2.336×10^{-3} , just 2.336

3b

b) The graph of $\frac{1}{T} \times 10^{-3}$ against $\ln \frac{1}{t}$ is:



- X-axis correctly labelled with $\frac{1}{T} \times 10^{-3}$

AND

Y-axis correctly labelled with $\ln \frac{1}{t}$; [1 mark]

- Appropriate scales selected for both axes; [1 mark]
- All points plotted ± 1 square; [1 mark]
- Straight line of best fit; [1 mark]

[Total: 4 marks]

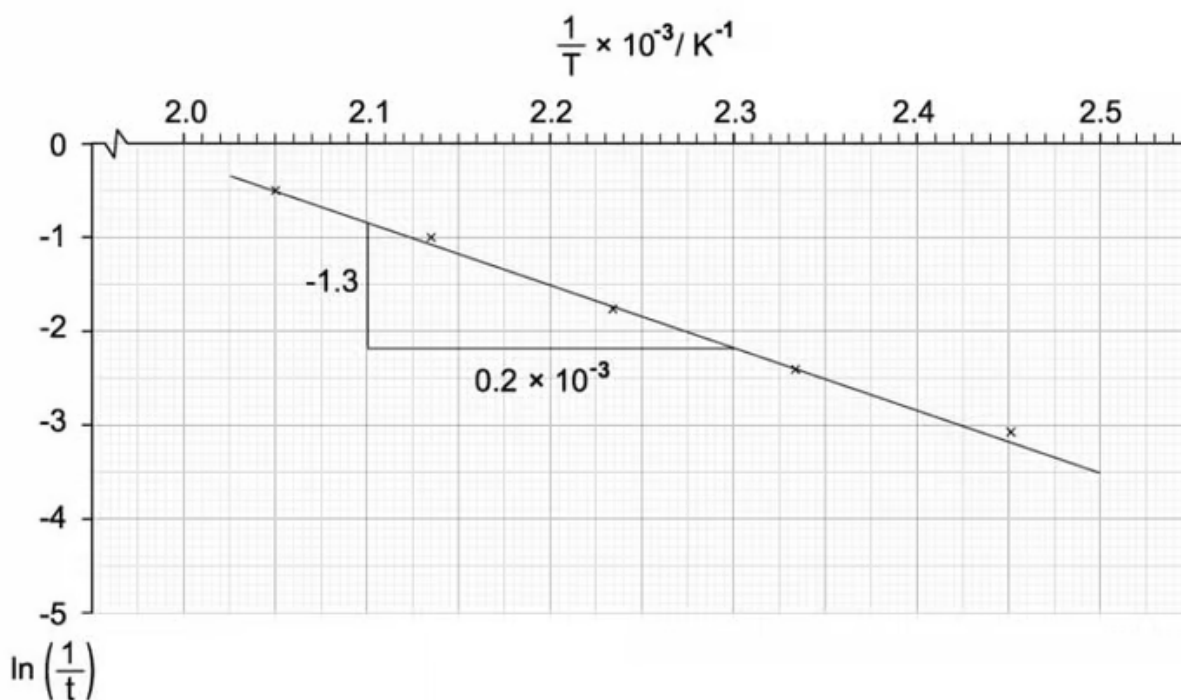
- Graphs associated with rate equation questions often have challenging scales

so take your time deciding on the direction and scales of the axes

- **Remember:** Your graph should take up at least half the space given as this can sometimes be a marking point

3c

c) To calculate a value for the activation energy, in kJ mol^{-1} , for this reaction:



- Calculate the gradient $= \frac{\Delta y}{\Delta x} = \frac{-E_a}{R} = \frac{-2.2 - (-0.9)}{2.3 \times 10^{-3} - 2.1 \times 10^{-3}} = \frac{-1.3}{0.2 \times 10^{-3}}$;
[1 mark]
- $= -6500$; [1 mark]
- $E_a = -(-6500) \times 8.31 = 54\,015 \text{ J mol}^{-1}$ [1 mark]
- $= 54.0 \text{ kJ mol}^{-1}$; [1 mark]

[Total: 4 marks]

- **Remember:** $y = mx + c$ applies to the logarithmic equation given in part (b)
 - Where gradient $= m = \frac{-E_a}{R}$, $x = \frac{1}{T}$, $c = \ln A$
- **Careful:** The x-axis scale is $\times 10^{-3}$ so don't forget that your value should be too
- Therefore, you need to $\times 10^{-3}$ to get the correct value
- When calculating E_a , remember to rearrange your $\frac{-E_a}{R}$ equation to give you just

$$F = -(\text{gradient value}) \times D$$